Stochastics - KW5

→ Stochastic process

Let $T \subseteq \mathbb{R}$ and $E \subseteq \mathbb{R}$. A stochastic process is a collection of random variables $(X_t)_{t \in T}$ with values in E.

(T: index set, E: state space)

For each elementary event ω, we have a map

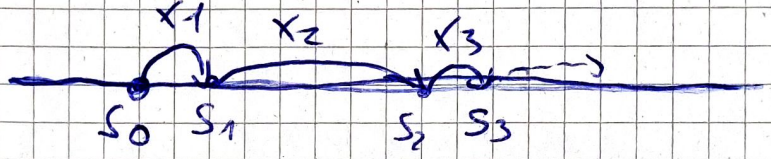
$$\underbrace{T \ni t \mapsto X_t(\omega)}_{\text{path}}$$

→ eg: S.z.

For i.i.d. random variables $(X_i)_{i \in \mathbb{N}}$ define

$$S_0 = 0 \quad \text{and} \quad S_t = \sum_{i=1}^{t} X_i \quad \text{for } t \in \mathbb{N}$$

Then $(S_t)_{t \in \mathbb{N}_0}$ is a discrete-time stochastic process and is called a random walk



If $P(X_1 = 1) = P(X_1 = -1) = \frac{1}{2}$, we denote $(S_t)_{t \in \mathbb{N}_0}$ as simple walk

We say that a stochastic process $(X_t)_{t \in T}$ has independent increments if, for all $t_0, \dots, t_n \in T$ with $t_0 < t_1 < \dots < t_n$ and $n \in \mathbb{N}$

$$X_{t_1} - X_{t_0},\ X_{t_2} - X_{t_1},\ \dots,\ X_{t_n} - X_{t_{n-1}}$$

are independent.

Stochastics - KVB

→ 5.3:

A stochastic process $(B_t)_{t \geq 0}$ is called a Brownian motion if

(i) $P(B_0 = 0) = 1$

(ii) $P([0, \infty) \ni t \mapsto B_t \text{ is continuous}) = 1$

(iii) $(B_t)_{t \geq 0}$ has independent increments

(iv) For $s, t \geq 0$ with $s \leq t$,

$$B_t - B_s \sim \text{Normal}(0, \sqrt{t-s})$$

→ For $t \geq 0$, $B_t \sim \text{Normal}(0, \sqrt{t})$

→ Mean function & covariance function

Let $(X_t)_{t \in T}$ be a stochastic process. If $E[|X_t|] < \infty$ for all $t \in T$, the mean function of $(X_t)_{t \in T}$ is given by

$$\mu(t) = E[X_t]$$

for $t \in T$. If $E[X_t^2] < \infty$ for all $t \in T$, the covariance function of $(X_t)_{t \in T}$ is defined as

$$C(s,t) = \text{Cov}(X_s, X_t)$$

for $s, t \in T$

Stochastics - KVD

~> Poisson Process

For $\lambda > 0$ and i.i.d. $Exp(\lambda)$ ~ distributed random variables $(W_i)_{i \in \mathbb{N}}$ define

$$X_n = \sum_{i=1}^{n} W_i \text{ for } n \in \mathbb{N} \text{ and } N_t = \sum_{n=1}^{\infty} 1_{\{X_n \leq t\}} \text{ for } t \geq 0$$

The stochastic process $(N_t)_{t \geq 0}$ is called Poisson Process with intensity λ.

~> Geiger counter

~> For $s, t \geq 0$ with $s \leq t$, $N_t - N_s \sim$ Poisson$(\lambda(t-s))$

~> $(N_t)_{t \geq 0}$ has independent increments.

~> Markov Property:

Let $(X_t)_{t \in T}$ be a stochastic process with an at most countable state space E. We say that $(X_t)_{t \in T}$ has the Markov Property if

$$P(X_t = i \mid X_{t_n} = i_n, \dots, X_{t_1} = i_1) = P(X_t = i \mid X_{t_n} = i_n)$$

for all $i_1, \dots, i_n, i \in E$, $t_1, \dots, t_n, t \in T$ with $t_1 < \dots < t_n < t$ and $n \in \mathbb{N}$ such that

$$P(X_{t_n} = i_n, \dots, X_{t_1} = i_1) > 0.$$

We call $(X_t)_{t \in T}$ a Markov Chain if it satisfies the Markov Property.

Stochastic KW3

→ 5.15:

Let $(X_t)_{t \in \mathbb{N}_0}$ be a discrete-time Markov Chain with an at most countable state space E.
We call real numbers $(p_{ij})_{i,j \in E}$ transition properties of $(X_t)_{t \in \mathbb{N}_0}$ if

$$P(X_{t+1} = j \mid X_t = i) = p_{ij}$$

for all $t \in \mathbb{N}_0$ and $i, j \in E$ such that $P(X_t = i) > 0$

→ eg:

A simple random walk $(S_t)_{t \in \mathbb{N}_0}$ is a discrete time Markov Chain with transition probabilities.

$$p_{i,j} = \begin{cases} 1/2 & , j \in \{i-1, i+1\} \\ 0 & , \text{else} \end{cases}$$

for $i, j \in \mathbb{Z}$

→ 5.18:

Let $(X_t)_{t \in \mathbb{N}_0}$ be a discrete-time Markov Chain with an at most countable state space E and transition probabilities $(p_{ij})_{i,j \in E}$. Then, for $t \in \mathbb{N}_0$ and $j \in E$,

$$P(X_{t+1} = j) = \sum_{i \in E} p_{ij} P(X_t = i)$$

Stochastics - KW3

-> 5.19:

Let $(X_t)_{t \in \mathbb{N}_0}$ be a discrete-time Markov-Chain with state space $E = \{1, .., m\}$ for some $m \in \mathbb{N}$ and transition probabilities $(p_{ij})_{i,j \in \{1,..,m\}}$. Define the transition matrix

$$P = \begin{pmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & & \\ p_{m1} & \cdots & p_{mm} \end{pmatrix} \quad \text{and} \quad p(t) = (P(X_t = 1), .., P(X_t = m))$$

and

$$p(t+1) = p(t) \cdot P$$

-> eg: 5.20:

A machine is operable or inoperable. Daily states describe as Markov Chain. If its is operable, it gets inoperable with probability $\frac{1}{10}$. For inoperable to operable with probability $1/2$. At time zero the machine is operable.

$$\rightsquigarrow P = \begin{pmatrix} 9/10 & 1/10 \\ 1/2 & 1/2 \end{pmatrix}, \quad p(0) = (1 \; 0)$$

$$p(t) = (P(X_t = 1) \; P(X_t = 2))$$

$$\rightarrow p(t) = p(0) P^t$$

Stochastiek - KVB - Ex 10

(1)

Let X be a random variable with $E[X^2] < \infty$ and let μ and $\sigma > 0$ denote its expectation & standard deviation.

a) Show that $P(|X - \mu| \geq t\sigma) < \frac{1}{t^2}$ for any $t > 0$

$\sigma = \sqrt{\text{Var}[X]}$, $E[X] = \mu$

Chebyshev's:

$$P(|X - \mu| \geq t\sigma) = P(|X - E[X]| \geq t\sqrt{\text{Var}[X]}) \leq \frac{\text{Var}[X]}{(t\sqrt{\text{Var}[X]})^2} = \frac{1}{t^2}$$

b) Compute the bounds for $t = 1$, $t = \sqrt{2}$, $t = 2$ & $t = 5$

$$P(|X - \mu| \geq \sigma) \leq 1$$

$$P(|X - \mu| \geq \sqrt{2}\,\sigma) \leq \frac{1}{2}$$

$$P(|X - \mu| \geq 2\sigma) \leq \frac{1}{4}$$

$$P(|X - \mu| \geq 5\sigma) \leq \frac{1}{25}$$

c)

$$0{,}9\sigma \leq P(X \in (\mu - c, \mu + c)) = P(|X - \mu| < c)$$

$$= 1 - P(|X - \mu| \geq c)$$

$$\Leftrightarrow P(|X - \mu| \geq c) = 0{,}04$$

$$\hookrightarrow c = 5\sigma$$

Stochastik KVB Ex 10

(1)

a) $X \sim Normal(\mu, \sigma)$ and $P(\mu - 3\sigma < X < \mu + 3\sigma)$

$P(\mu - 3\sigma < X < \mu + 3\sigma) = P(|X - \mu| < 3\sigma)$

$= 1 - P(|X - \mu| \geq 3\sigma) =$

$P(-3\sigma < X - \mu < +3\sigma) = P(-3 < \frac{X-\mu}{\sigma} < 3)$

$= P\left(\frac{X-\mu}{\sigma} \leq 3\right) - P\left(\frac{X-\mu}{\sigma} \leq -3\right)$

$= \Phi(3) - \Phi(-3) = \Phi(3) - (1 - \Phi(3)) = 0,9974$

$\Phi(-t) = 1 - \Phi(t)$ $\Leftarrow (=) \Rightarrow$ bei $Normal(0,1)$

$X \sim Normal(\mu, \sigma)$

$X - \mu \sim Normal(0, \sigma)$

$\frac{X-\mu}{\sigma} \sim Normal(0, 1)$

Stochastik – Ex 10 – KW 20

② Let $(Z_t)_{t\in\mathbb{N}}$ be a sequence of i.i.d. random variables with $E[Z_1]=\mu$ & $Var[Z_1]=\sigma^2$. The process $(X_t)_{t\in\mathbb{Z}}$ is given by $X_t = \frac{Z_{t-1}+Z_t+Z_{t+1}}{3}$ for $t\in\mathbb{Z}$

a) Calculate the mean function $\mu(t)=E[X_t]$ of $(X_t)_{t\in\mathbb{Z}}$

$$\mu(t)=E[X_t] = E\left[\frac{Z_{t-1}+Z_t+Z_{t+1}}{3}\right] = \frac{1}{3}E[Z_{t-1}+Z_t+Z_{t+1}]$$

$$= \frac{3}{3}E[Z_t] = \mu$$

b) Determine the covariance function $\rho(s,t)=Cov(X_s,X_t)$ of $(X_t)_{t\in\mathbb{Z}}$

$$\rho(s,t) = Cov(X_s,X_t) = E[X_s X_t] - E[X_s]E[X_t]$$

$$= E\left[\left(\frac{Z_{t-1}+Z_t+Z_{t+1}}{3}\right)\left(\frac{Z_{s-1}+Z_s+Z_{s+1}}{3}\right)\right] - \mu\cdot\mu \qquad u\in\mathbb{Z}, t\in\mathbb{Z}$$

$$Cov(Z_u, X_t) = Cov\left(Z_u, \frac{Z_{t-1}+Z_t+Z_{t+1}}{3}\right)$$

$$= \frac{1}{3}\left(Cov(Z_u,Z_{t-1}) + Cov(Z_u,Z_t) + Cov(Z_u,Z_{t+1})\right)$$

($Cov(Z_u,Z_{t-1})$: if same then $= Var[Z_t]$)

$$= \begin{cases} \frac{1}{3}\sigma^2, & u\in\{t-1,t,t+1\} \\ 0, & \text{else} \end{cases}$$

Stochastics - KVB - 6x90

(2)

b)

$$\operatorname{cov}(X_s, X_t) = \operatorname{cov}\left(\frac{Z_{s-1}+Z_s+Z_{s+1}}{3}, X_t\right)$$

$$= \frac{1}{3}\left(\operatorname{cov}(Z_{s-1}, X_t) + \operatorname{cov}(Z_s, X_t) + \operatorname{cov}(Z_{s+1}, X_t)\right)$$

$$= \begin{cases} \frac{1}{3}\sigma^2 & , s = t \\ \frac{2}{9}\sigma^2 & , s \in \{t-1, t+1\} \\ \frac{1}{9}\sigma^2 & , s \in \{t-2, t+2\} \\ 0 & , \text{else} \end{cases}$$

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Stochastics – KVB – Ex 10

(3)

Let $(N_t)_{t\geq 0}$ be a Poisson Process with intensity $\lambda > 0$.
Determine the conditional probabilities $P(N_s = k \mid N_t = n)$
for $0 < s < t$ and $0 \leq k \leq n$.
Hint: $P(N_0 = 0) = 1$ and that the increments of a Poisson
process are independent and Poisson distributed with parameter
$\lambda(t-s)$ for $0 \leq s \leq t$

$$P(N_s = k \mid N_t = n) = \frac{P(N_s = k, N_t = n)}{P(N_t = n)}$$

$$= \frac{P(N_s = k, N_t - N_s = n-k)}{P(N_t = n)}$$ — in Form für (ii) bringen

$N_0 = 0$ (i)

$$= \frac{P(N_s - N_0 = k-0,\ N_t - N_s = n-k)}{P(N_t = n)}$$

independent (ii)

$$= \frac{P(N_s - N_0 = k-0)\, P(N_t - N_s = n-k)}{P(N_t = n)}$$

distributed (i.i.)

$$= \frac{\frac{\lambda^k (s-0)^k}{k!} e^{-\lambda s} \cdot \frac{\lambda^{n-k} (t-s)^{n-k}}{(n-k)!} e^{-\lambda(t-s)}}{\frac{\lambda^n t^n}{n!} e^{-\lambda t}}$$

$$= \frac{n!}{k!\,(n-k)!} \left(\frac{s}{t}\right)^k \left(\frac{t-s}{t}\right)^{n-k} = \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k}$$

Stochastics – KVB – Ex 10

(1)

Let $(B_t)_{t \geq 0}$ be a Brownian motion

a) Show that $(-B_t)_{t \geq 0}$ is a Brownian Motion

(i) $P(B_0 = 0) = 1$ is still correct $P(-B_0 = 0) = P(B_0 = 0) = 1$

(ii) $P([0,\infty) \ni t \mapsto B_t$ is continuous$) = 1$ Shaking does not affect continuous of function

(iii) For $0 < t_0 < \dots < t_n$,

$B_{t_1} - B_{t_0}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent

$= -(-B_{t_1} - (-B_{t_0})), \dots, -(-B_{t_n} - (-B_{t_{n-1}}))$

so that $-B_{t_1} - (-B_{t_0}), \dots$ are independent!

(iv) For $0 \leq s \leq t$

$$-B_t - (-B_s) = \underbrace{-(B_t - B_s)}_{\sim \text{Normal}((0, \sqrt{t-s})} \sim \text{Normal}(0, \sqrt{t-s})$$

$\Rightarrow$ Brownian Motion

(4)

5) Let $c>0$. Show that $\left(\frac{1}{\sqrt{c}} B(ct)\right)_{t\geq 0}$ is a Brownian Motion

(i) $P(B_0 = 0) = 1 = P\left(\frac{1}{\sqrt{c}} B_{c0} = 0\right)$ ✓

(ii) scaling does not affect continuously. ✓

(iii) For $0 < t_0 < \dots < t_n$,

$B_{t_1} - B_{t_0}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent

$= \sqrt{c}\left(\frac{1}{\sqrt{c}} B_{t_1} - \frac{1}{\sqrt{c}} B_{t_0}\right), \dots, \sqrt{c}\left(\frac{1}{\sqrt{c}} B_{t_n} - \frac{1}{\sqrt{c}} B_{t_{n-1}}\right)$

$\rightsquigarrow \frac{1}{\sqrt{c}} B_{t_1} - \frac{1}{\sqrt{c}} B_{t_0}, \dots$ are independent

(iv) For $0 \leq s \leq t$

$B_{ct} - B_{cs} \sim \text{Normal}(0, \sqrt{c(t-s)})$

so that $\frac{1}{\sqrt{c}}(B_{ct} - B_{cs}) \sim \text{Normal}(0, \sqrt{t-s})$

Thus $\left(\frac{1}{\sqrt{c}} B_{ct}\right)_{t\geq 0}$ is Brownian Motion

Stochastics - KVB - Ex 16

(9)

a) Let $\mu \in \mathbb{R}$, $\sigma > 0$ and $(X_t)_{t \geq 0}$ given by $X_t = \mu t + \sigma B_t$ for all $t \geq 0$. Determine the distribution of the increments $X_t - X_s$ for $0 \leq s \leq t$ and sketch $(X_t)_{t \geq 0}$

$$X_t = \mu t + \sigma B_t$$

$$B_t - B_s \sim \text{Normal}((0, \sqrt{t-s}))$$

$$B_t \sim \text{Normal}((0, \sqrt{t}))$$

$$X_t = \mu t + \sigma B_t \sim \text{Normal}((\mu t, \sigma\sqrt{t}))$$

$$X_t - X_s = \text{Normal}((\mu t - \mu s, \sigma\sqrt{(t-s)}))$$

Stochastics - KVB

→ Basic Notion of Statistics

Observations $x_1, \dots, x_n \in \mathbb{R}$ from a sample $x = (x_1, \dots, x_n) \in \mathbb{R}^n$

→ 6.1.

For a sample $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ define the sample mean $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ and

the sample variance

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

if $n \geq 2$. For $i \in \{1, \dots, n\}$ let $x_{(i)}$ be the ith smallest value of x and define the sample median

$$m_x = \begin{cases} x\left(\frac{(n+1)}{2}\right), & n \text{ is odd} \\ \frac{x\left(\frac{n}{2}\right) + x\left(\frac{n}{2}+1\right)}{2}, & n \text{ is even} \end{cases}$$

m_x is median if $P(X \geq m) \geq \frac{1}{2}$ & $P(X \leq m) \geq \frac{1}{2}$

→ Statistical Model:

A statistical model $(X, F_X, (P_\vartheta)_{\vartheta \in \Theta})$ consist of a sample space X, a σ-algebra F_X on X and a family of probability measures $(P_\vartheta)_{\vartheta \in \Theta}$ on X with an index set Θ.

→ X possible values of sample $x = (x_1, \dots, x_n)$

→ For each $\vartheta \in \Theta$, (X, F_X, P_ϑ) is a probability space

→ Ideas: $x_1, \dots, x_n$ are realisations of random variables $X_1, \dots, X_n$ with joint distribution P_ϑ for some „true" but unknown $\vartheta \in \Theta$.

Stochastik - KVB - C

→ e.g.: 6.4:

For n repetitions of a random experiment with success and failure (1|0) and unknown success probability p the statistical model is $([0,1]^n, P([0,1]^n), (P_p)_{p\in[0,1]})$ where P_p is the joint distribution of n independent random variables $Bea(p) \sim$ distributed $X_1, \dots, X_n$ for $p \in [0,1]$

→ Statistical procedures:

Sample x from P_ϑ for unknown $\vartheta \in \Theta$

- Point estimation: guess for ϑ or $\tau(\vartheta)$
- Confidence regions: set that contains ϑ or $\tau(\vartheta)$ with some certainty
- Statistical test: For $\Theta_0, \Theta_1 \subseteq \Theta$ with $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$ decide with some certainty $\vartheta \in \Theta_0$ or $\vartheta \in \Theta_1$

Stochastics - KVB -

→ Point Estimator

$(X, \mathcal{F}_X, (P_\vartheta)_{\vartheta \in \Theta})$ statistical model, $\tau : \Theta \to \mathbb{R}$

Sample X from P_ϑ for some unknown $\vartheta \in \Theta$.

Goal: estimate $\tau(\vartheta)$

→ Def 6.6:

A measurable map $T : \mathcal{X} \to \mathbb{R}$ is called a point estimator

→ 6.7:

→ For each $\vartheta \in \Theta$, T is a random variable on $(X, \mathcal{F}_X, P_\vartheta)$

→ eg: sample mean

→ Unbiased estimator and bias of an estimator

Let T be an estimator such that $E_\vartheta[|T|] < \infty$ for all $\vartheta \in \Theta$. We call T an unbiased estimator for $\tau(\vartheta)$ if

$$E_\vartheta[T] = \tau(\vartheta) \quad \text{for all } \vartheta \in \Theta$$

Otherwise, we denote $B_T(\vartheta) = E_\vartheta[T] - \tau(\vartheta)$ as bias of T at $\vartheta \in \Theta$

Stochastics - KVB

6.9.

Let $(X, \mathcal{F}_X, (P_\vartheta)_{\vartheta \in \Theta})$ be a statistical model such that P_ϑ is the distribution of $X = (X_1, \dots, X_n)$ with i.i.d. random variables $X_1, \dots, X_n$ for $\vartheta \in \Theta$

(i) Assume that $E_\vartheta[|X_1|] < \infty$ for all $\vartheta \in \Theta$. Then, the sample mean $\overline{X}$ is an unbiased estimator for $E_\vartheta[X_1]$,

$$E_\vartheta[\overline{X}] = E_\vartheta[X_1] \quad \text{for all } \vartheta \in \Theta$$

(ii) Assume that $E_\vartheta[X_1^2] < \infty$ for all $\vartheta \in \Theta$ and that $n \geq 2$. Then, the sample variance S_X^2 is an unbiased estimator for $\mathrm{Var}_\vartheta[X_1]$.

$$E_\vartheta[S_X^2] = \mathrm{Var}_\vartheta[X_1] \quad \text{for all } \vartheta \in \Theta$$

Consistency:

Let $(X, \mathcal{F}_X, (P_\vartheta)_{\vartheta \in \Theta})$ be a statistical model such that P_ϑ is the joint distribution of random variables $(X_i)_{i \in \mathbb{N}}$ for each $\vartheta \in \Theta$. Let $(T_n)_{n \in \mathbb{N}}$ be a sequence of estimators such T_n takes only the realisations of $X_1, \dots, X_n$ into account. We say that the estimators $(T_n)_{n \in \mathbb{N}}$ are consistent for $\tau(\vartheta)$ if for all $\vartheta \in \Theta$ and $\varepsilon > 0$,

$$\lim_{n \to \infty} P_\vartheta(|T_n - \tau(\vartheta)| \geq \varepsilon) = 0,$$

if $T_n \xrightarrow{P_\vartheta} \tau(\vartheta)$ as $n \to \infty$ for all $\vartheta \in \Theta$

Stochastik KVB

↝ eg: 6.12.

Let $(X, F_X, (P_\vartheta)_{\vartheta \in \Theta})$ be a statistical model such that P_ϑ is the joint distribution of i.i.d. random variables $(X_i)_{i \in \mathbb{N}}$ for each $\vartheta \in \Theta$. For $n \in \mathbb{N}$ let T_n be the sample mean of $X_1, \dots, X_n$. By the weak law of large numbers,

$$T_n = \frac{1}{n} \sum_{i=1}^{n} X_i \xrightarrow{P_\vartheta} E_\vartheta[X_1] \text{ as } n \to \infty$$

for all $\vartheta \in \Theta$, whence the estimators $(T_n)_{n \in \mathbb{N}}$ are consistent for $E_\vartheta[X_1]$

↝ 6.13:

A further possibility to measure the imprecision of an estimator T for $\tau(\vartheta)$ is the mean squared error

$$E_\vartheta[(T - \tau(\vartheta))^2] \quad \text{for } \vartheta \in \Theta$$

If $E_\vartheta[T^2] < \infty$ for all $\vartheta \in \Theta$, we have

$$E_\vartheta[(T - \tau(\vartheta))^2] = \mathrm{Var}_\vartheta[T] + B_T(\vartheta)^2$$

for $\vartheta \in \Theta$, where $B_T(\vartheta)$ is the bias of T at ϑ

Stochastics IKVB

→ 6.148

Let $(X, P(X), (P_\vartheta)_{\vartheta \in \Theta})$ be a statistical model with X at most countable and $\Theta \subseteq \mathbb{R}$. We call $T: X \to \Theta$ a maximum likelihood estimator for ϑ if

$$P_{T(x)}(\{x\}) = \max_{\vartheta \in \Theta} P_\vartheta(\{x\})$$

for all $x \in X$, $T(x)$ is a maximiser of $\alpha_x : \Theta \ni \vartheta \mapsto P_\vartheta(\{x\})$ for all $x \in X$

The idea of maximum likelihood estimation is to choose that ϑ from Θ with the highest probability for the observed sample

e.g. → 6.16:

Let $(\mathbb{N}, P(\mathbb{N}), (P_m)_{m \in \mathbb{N}})$ be a statistical Model, where P_m is such that $P_m(\{i\}) = \frac{1}{m}$ for $i \in \{1, \dots, m\}$ for $m \in \mathbb{N}$. For $x \in \mathbb{N}$ and $m \in \mathbb{N}$,

$$P_m(\{x\}) = \begin{cases} 1/m & , x \leq m \\ 0 & , x > m \end{cases}$$

Thus, $m \mapsto P_m(\{x\})$ is maximal for $m = x$ and $T(x) = x$ is the maximum likelihood estimator for m.

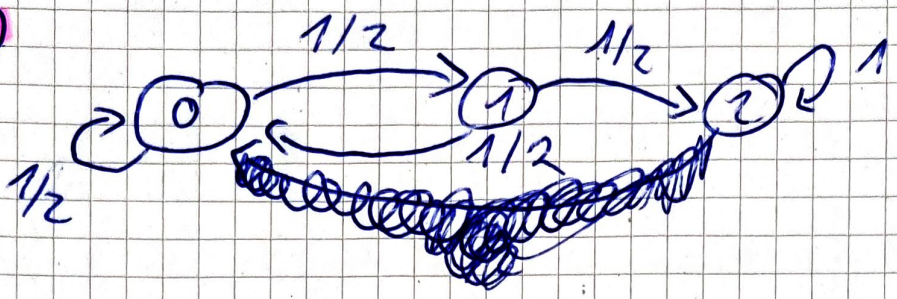
Stochastik - KV 3 - Ex 11

a) $(X_t)_{t \in \mathbb{N}}$ with

$X_t : \{\text{'tails'}, \text{'heads'}\} \to \{0,1,2\}$

for $t \in \mathbb{N}_0$. Here X_t is the number of successive heads at time t until we have two successive heads

b)



c) $$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}$$

d) $p(t) = (P(X_t=0)\ P(X_t=1)\ P(X_t=2))$

$p(0) = (1\ 0\ 0)$

$$p(1) = (1\ 0\ 0)\begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 0 & 1 \end{pmatrix} = (1/2 \quad 1/2 \quad 0)$$

$$p(2) = (1/2\ 1/2\ 0)\begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 0 & 1 \end{pmatrix} = \left(\frac{1}{2} \quad \frac{1}{4} \quad \frac{1}{4}\right)$$

e)

$$\lim_{t \to \infty} p(t) = (0\ 0\ 1)$$

Stochastik - KVB - Ex 11

(2) Let $(\mathcal{X}, \mathcal{F}_{\mathcal{X}}, (P_m)_{m \in \mathbb{R}})$ be a statistical model, where P_m is the joint distribution of a sequence of iid random variables $(X_i)_{i \in \mathbb{N}}$ with $E_m[X^2] < \infty$ and $E_m[X_1] = m$. For $n \geq 2$ the sample $x_1, \dots, x_n$ is given.

Let $\hat{x}_n = \underbrace{\frac{1}{n-1} \sum_{i=1}^{n} x_i}_{\hat{m}}$ be an estimator for the mean m.

a) $\tau(\vartheta) = m$

$$B_{\hat{x}_n}(m) = E_{(m)}[\hat{x}_n] - m = E_m\left[\frac{1}{n-1} \sum_{i=1}^{n} x_i\right] - m$$

$$= \frac{1}{n-1} \cdot n \cdot m - m = \frac{m}{n-1} \Rightarrow \text{Biased!}$$

For $n \to \infty$:

$$B_{\hat{x}_n}(m) \Rightarrow 0$$

b) Is $(\hat{x}_n)_{n \geq 2}$ consistent?

We have ($\alpha \propto N$)

$$\frac{1}{n} \sum_{i=1}^{n} x_i \xrightarrow{P} E[X_1] = m$$

It holds that

$$\hat{x}_n = \frac{1}{n-1} \sum_{i=1}^{n} x_i = \underbrace{\frac{n}{n-1}}_{\to 1} \cdot \underbrace{\frac{1}{n} \sum_{i=1}^{n} x_i}_{m} \xrightarrow{P} m$$

consistent!

$\forall \varepsilon > 0 : P(|\hat{x}_n - m| > \varepsilon) \to 0$

Stochastik - KVB - Ex 11

①

Let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (P_a)_{a>0})$ be a statistical model, where P_a is the joint distribution of n independent uniformly distributed random variables $U_1, \dots, U_n$ on $[0, a]$. Let $u = (u_1, \dots, u_n)$ be a given sample. In this task

a) Let $\hat{a}_1 = 2\bar{u}$ be an estimate of a.

$$\bar{u} = \frac{1}{n} \sum_{i=1}^{n} u_i$$

$$B_{\hat{a}_1}(a) = E_a[2\bar{U}] - a = E_a\left[\frac{2}{n} \sum_{i=1}^{n} U_i\right] - a$$

$$= \frac{2}{n} E_a[U_1] - a = \frac{2}{2} a - a = 0$$ unbiased

b) Let $\hat{a}_2 = \max_{i \in \{1,\dots,n\}} u_i$ be an estimate of a.

Density $f(x) = \begin{cases} \frac{n}{a^n} x^{n-1}, & x \in [0, a] \\ 0, & \text{else} \end{cases}$

Fixed $t \in \mathbb{R}$:

$$P_a(\hat{a}_2 \leq t) = P_a\left(\max_{i=(1,\dots,n)} u_i \leq t\right)$$

$$= P_a(u_1 \leq t, \dots, u_n \leq t) = P_a(u_1 \leq t) \cdot P_a(u_n \leq t)$$

$$= P_a(u_1 \leq t)^n$$

$$= \begin{cases} 0, & t \leq 0 \\ \left(\frac{t}{a}\right)^n, & 0 < t < a \\ 1, & t \geq a \end{cases} = F_X(x)$$

$$F_X(x)' = f(x) = \begin{cases} \frac{n x^{n-1}}{a^n}, & x \in [0, a] \\ 0, & \text{else} \end{cases}$$

Skoda - KVB - C4H

3)

5) $E_a[\hat{a}_2] = \int_{-\infty}^{\infty} x \cdot f(x)\,dx = \int_0^a x \cdot \frac{n}{a^n} \cdot x^{n-1}\,dx$

$= \int_0^a \frac{n x^n}{a^n}\,dx = \left[\frac{n x^{n+1}}{(n+1)a^n}\right]_0^a = \frac{n a^{n+1}}{(n+1)a^n} = \frac{an}{n+1}$

$B_{\hat{a}_2}(a) = E_a[\hat{a}_2] - a = \frac{an}{n+1} - a \cdot \frac{n+1}{n+1} = \frac{a(n-n+1)}{n+1}$

$= \frac{a}{n+1}$ Drake :(

$\hat{a}_3 = c\hat{a}_2$

$E_a[c\hat{a}_2] = c \cdot \frac{an}{n+1} \overset{!}{=} a$

$\rightsquigarrow c = \frac{n+1}{n}$

Slenkis 6x11 — KUB

(3)

a) Variancias $\hat{a}_3$ ir $\hat{a}_1$.

$$Var_a[\hat{a}_1] = Var_a\left[\frac{2}{n}\sum_{i=1}^{n} a_i\right] \overset{ind.}{=} \frac{4}{n^2}\sum_{i=1}^{n} Var_a[a_i]$$

$$= \frac{a^3}{3n}$$

$$Var_a[\hat{a}_3] = E_a[\hat{a}_3^2] - E_a[\hat{a}_3]^2 = \frac{a^2}{n(n+2)}$$

$$E_a[\hat{a}_3^2] = \int_0^a \epsilon^2 \frac{n\epsilon^{n-1}}{a^n}\, d\epsilon$$

Which is better

$\hat{a}_3$, since $Var_a[\hat{a}_3] \leq Var_a[\hat{a}_1]$

Stochastic - KVB - Ex 11

(9)

Let $(\mathbb{N}_0^n, P(\mathbb{N}_0^n), (P_p)_{p\in(0,1)})$ be a statistical model, where P_p is the joint distribution of n geometrically distributed r.V.s $X_1, \dots, X_n$ with parameter $p \in (0,1)$. Let $x = (x_1, \dots, x_n) \in \mathbb{N}_0^n$ be a given sample

a) Calculate the likelihood function $\mathcal{L}_x(p) = P_p(\{x\})$

$$\mathcal{L}_x(p) = P_p(\{x\}) = P_p(X_1 = x_1, \dots, X_n = x_n)$$

$$\overset{\text{indep.}}{=} P_p(X_1 = x_1) \cdot \dots \cdot P_p(X_n = x_n)$$

$$= \prod_{i=1}^{n} P_p(X_i = x_i) = \prod_{i=1}^{n} p^{x_i}(1-p)$$

$$\hookrightarrow P(X = q) = p^q(1-p)$$

Nicht Max nur Funktion aufstellen ok

Skript 18/B-6x11

(a)

b) Determine the maximum likelihood estimator for p

$$\log\left(\mathcal{L}_X(p)\right) = \log\left(\prod_{i=1}^{n} p^{x_i}(1-p)\right) = \sum_{i=1}^{n} \log\left(p^{x_i}(1-p)\right)$$

$$= \sum_{i=1}^{n} \left(\log\left(p^{x_i}\right) + \log(1-p)\right)$$

$$= \sum_{i=1}^{n} x_i \log(p) + n \cdot \log(1-p)$$

$$\sum_{i=1}^{n} x_i = \bar{x} \cdot n$$

$$= n \cdot \bar{x} \cdot \log(p) + n \cdot \log(1-p)$$

$$\frac{d}{dp} \log\left(\mathcal{L}_X(p)\right) = \frac{n\bar{x}}{p} - \frac{n}{1-p} \overset{!}{=} 0$$

$$\Leftrightarrow p = \frac{\bar{x}}{1+\bar{x}}$$

$$\rightsquigarrow \hat{p} = \frac{\bar{x}}{1+\bar{x}} \text{ is the MLE}$$

Statistik KW3-Ex 11

(5)

There is a population of fish in a pond. To estimate the population size N, $M \leq N$ fishes are caught, marked and released. After a certain time, $n \leq M$ fishes are caught again and the marked ones are counted to estimate the population size N.

a) Statistical Model

N fishes / M marked, $N-M$ unmarked

We catch n fishes

M marked, $N-M$ unmarked.

x - Number of marked fishes

$$P_N(X=a) = \frac{\binom{M}{a}\binom{N-M}{n-a}}{\binom{N}{n}} \qquad a \in \{0,\dots,n\}$$

$$\left(\{0,\dots,n\}, P(\{0,\dots,n\}), (P_N)_{N\in\{M,\dots\}}\right)$$

b) Let $x \in \{0,\dots,n\}$ be a given sample. Show that $\frac{x}{Mn}$ is unbiased for $\frac{1}{N}$

$$\text{Bias}_{\frac{x}{Mn}}\left[\frac{1}{N}\right] = E_{1/N}\left[\frac{x}{Mn}\right] - \frac{1}{N} = 0$$

$$E_{\frac{1}{N}}\left[\frac{x}{Mn}\right] = \sum_{a=0}^{n} \frac{a}{Mn} \cdot \frac{\binom{M}{a}\binom{N-M}{n-a}}{\binom{N}{n}} = \sum_{a=1}^{n} \frac{a\frac{M}{a}\binom{M-1}{a-1}\binom{N-M}{n-a}}{\frac{N}{n}\binom{N-1}{n-1}}$$

$$= \frac{Mn}{N}\sum_{a=1}^{n} \frac{\binom{M-1}{a-1}\binom{N-M}{n-a}}{\binom{N-1}{n-1}} = \frac{Mn}{N}\sum_{a=0}^{n-1} \frac{\binom{M-1}{a}\binom{N-M}{n-(a+1)}}{\binom{N-1}{a-1}} = \frac{Mn}{N}\frac{1}{n}$$

Stochastics - KVB

→ Confidence Intervals

Confidence region: set that contains $\tau(\vartheta)$ with some certainty

I set of intervals on $\mathbb{R}$

→ Def 6.19:

Let $(X, \mathcal{F}_X, (P_\vartheta)_{\vartheta \in \Theta})$ be a statistical model and let $\tau : \Theta \to \mathbb{R}$. We call $C : X \to \mathcal{I}$ a confidence interval for $\tau(\vartheta)$ with error level $\alpha \in (0,1)$ (or confidence level $1-\alpha$) if

$$P_\vartheta(\{x \in X : \tau(\vartheta) \in C(x)\}) \geq 1 - \alpha$$

for all $\vartheta \in \Theta$.

→ eg: 6.20

For $n \in \mathbb{N}$ and $\sigma_0 > 0$ let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (P_\mu)_{\mu \in \mathbb{R}})$ be the statistical model such that P_μ is the joint distribution of independent $X_1, \dots, X_n \sim \text{Normal}(\mu, \sigma_0)$ for $\mu \in \mathbb{R}$. Let $X = (X_1, \dots, X_n)$.

For $\alpha \in (0,1)$ choose $z_{1-\alpha/2} \in \mathbb{R}$ with $\Phi(z_{1-\alpha/2}) = 1 - \alpha/2$

Define: $C : \mathbb{R}^n \ni x \mapsto \left(\bar{x} - z_{1-\alpha/2} \frac{\sigma_0}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2} \frac{\sigma_0}{\sqrt{n}}\right)$

For any $\mu \in \mathbb{R}$, $P_\mu(\{x \in X : \mu \in C(x)\}) =$

$$\Leftrightarrow P_\mu\left(\{x \in X, \mu \in \left(\bar{x} - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right)\}\right)$$

$$\bar{x} \in \left(\mu - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \mu + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right)$$

$$= P_\mu\left(\bar{x} \in \left(\mu - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \mu + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right)\right)$$

(Sum E) (Sum V)

$$\sum_{i=1}^n X_i \sim \text{Normal}(n\mu, \sqrt{n}\,\sigma_0) \Rightarrow \frac{1}{n}\sum_{i=1}^n X_i \sim \text{Nord}\left(\mu, \frac{\sigma_0}{\sqrt{n}}\right)$$

Stochastics - KVB

→ Eg. 6.20:

$$= P_\mu\left(\frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \in (-z_{1-\alpha/2}, z_{1-\alpha/2})\right)$$

$\frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \sim \text{Normal}(0,1)$

$$= \phi(z_{1-\alpha/2}) - \phi(-z_{1-\alpha/2}) = \phi(z_{1-\alpha/2}) - (1-\phi(z_{1-\alpha/2}))$$
$$= 2 - \alpha - 1 = 1-\alpha$$

Thus, C is a confidence interval for μ with error level α.

→ Quantile:

For a random variable X and $p \in (0,1)$, $q_p \in \mathbb{R}$ is called a p-quantile of X if

$$P(X \leq q_p) \geq p \quad \& \quad P(X \geq q_p) \geq 1-p$$

For $p \in (0,1)$ the unique p-quantile z_p of the standard normal distribution solves

$$\phi(z_p) = p$$

Stochastics - KVB

⇝ Statistical Tests

⇝ $(X, F_X, (P_\vartheta)_{\vartheta \in \Theta})$ statistical model

⇝ $\Theta_0, \Theta_1 \subset \Theta$ such that $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$

⇝ Sample x from P_ϑ with unknown $\vartheta \in \Theta$

⇝ Decide if

$$\underbrace{H_0 : \vartheta \in \Theta_0}_{\text{null Hypothesis}} \quad \text{or} \quad \underbrace{H_1 : \vartheta \in \Theta_1}_{\text{alternative}}$$

⇝ eg:

Statistical Model $([0,1]^n, \mathcal{P}([0,1]^n), (P_p)_{p \in [0,1]})$

where P_p is the joint distribution of independent $X_1, \dots, X_n \sim \text{Ber}(p)$ for $p \in [0,1]$.

Decide if $H_1 : p > 0{,}5$ or $H_0 : p \leq 0{,}5$

⇝ Test

A measurable map $\varphi : X \to [0,1]$ is called a test.

If $\varphi(x) = 1$, we reject H_0 in favour of H_1.

If $\varphi(x) = 0$, we cannot reject H_0.

	$\vartheta \in \Theta_0$	$\vartheta \in \Theta_1$
$\varphi(x) = 0$	correct	type II Error ← Null H_0 is wrong but not rejected
$\varphi(x) = 1$	type I Error ← Null H_0 is true but rejected	correct

Stochastics - KVB

→ Significance level and power of a test

Let $\varphi : \mathcal{X} \to \{0,1\}$ be a test for $H_0 : \vartheta \in \Theta_0$ against $H_1 : \vartheta \in \Theta_1$

(i) We say that φ is a test of (significance) level α if

$$P_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\}) \leq \alpha$$

for all $\vartheta \in \Theta_0$

(ii) For $\vartheta \in \Theta_1$ we call

$$P_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\})$$

the power of φ at ϑ.

Idea: For $S : \mathcal{X} \to \mathbb{R}$ and $K \subseteq \mathbb{R}$

$$\varphi(x) = \begin{cases} 0, & S(x) \notin K \\ 1, & S(x) \in K \end{cases}$$

Stochastik - KVB

→ eg: (One-sided Binomial test)

Statistical model $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (P_p)_{p\in[0,1]})$ where P_p is the joint distribution of independent $X_1,\dots,X_n \sim Ber(p)$ for $p \in [0,1]$

Goal: Test with level $\alpha \in (0,1)$ for

$$H_0: p \le p_0 \quad \text{against} \quad H_1: p > p_0 \qquad (*)$$

for some $p_0 \in (0,1)$.

Let $m_0 \in \mathbb{N}_0$ be the smallest $m \in \mathbb{N}_0$ such that

$$\sum_{k=m+1}^{n} \binom{n}{k} p_0^k (1-p_0)^{n-k} \le \alpha$$

Define $\varphi: \{0,1\}^n \ni x = (x_1,\dots,x_n) \mapsto \begin{cases} 0, & \sum_{i=1}^n x_i \le m_0 \\ 1, & \sum_{i=1}^n x_i > m_0 \end{cases}$

For $p \in [0,1]$ with $p \le p_0$:

$$P_p(\{x \in \{0,1\}^n : \varphi(x) = 1\}) = P_p(\{x \in \{0,1\}^n : \sum_{i=1}^n x_i > m_0\})$$

$$= P_p(\underbrace{\sum_{i=1}^n x_i}_{\sim \text{Binomial}(n,p)} > m_0) \le P_{p_0}(\sum_{i=1}^n x_i > m_0) \le \alpha$$

Thus, φ is a level-α test for $(*)$

Stochastics - KVB

→ **Eg:** (One-sided Gauss test)

For $n \in \mathbb{N}$ and $\sigma_0 > 0$ let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (P_\mu)_{\mu \in \mathbb{R}})$ be a statistical model, where P_μ is the joint distribution of independent $X_1, \dots, X_n \sim \text{Normal}(\mu, \sigma_0)$.

Let $X = (X_1, \dots, X_n)$

$$H_0 : \mu \leq \mu_0 \text{ against } H_1 : \mu \geq \mu_0 \quad (**)$$

for $\mu_0 \in \mathbb{R}$. For $\alpha \in (0,1)$ let

$$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \bar{x} < \mu_0 + z_{1-\alpha} \frac{\sigma_0}{\sqrt{n}} \\ 1, & \bar{x} \geq \mu_0 + z_{1-\alpha} \frac{\sigma_0}{\sqrt{n}} \end{cases}$$

For $\mu \in \mathbb{R}$ with $\mu \leq \mu_0$,

$$P_\mu(\{x \in \mathbb{R}^n : \varphi(x) = 1\}) = P_\mu(\{x \in \mathbb{R}^n : \bar{x} \geq \mu_0 + z_{1-\alpha} \frac{\sigma_0}{\sqrt{n}}\})$$

$$= P_\mu\left(\bar{X} \geq \mu_0 + z_{1-\alpha} \frac{\sigma_0}{\sqrt{n}}\right)$$

$\bar{X} \sim \text{Normal}(\mu, \sigma_0/\sqrt{n})$

$$= P_\mu\left(\frac{\bar{X} - \mu}{\sigma_0/\sqrt{n}} \geq \frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} + z_{1-\alpha}\right) \leq P_\mu\left(\frac{\bar{X} - \mu}{\sigma_0/\sqrt{n}} \geq z_{1-\alpha}\right)$$

$$= 1 - P_\mu\left(\frac{\bar{X} - \mu}{\sigma_0/\sqrt{n}} < z_{1-\alpha}\right) = 1 - \Phi(z_{1-\alpha}) = \alpha$$

Thus, φ is a level-α test for $(**)$

Statistik - KW 3

→ eg.: (Two-sided Gauss test)

Let the statistical model as in eg. 6.3.1 be.
Test

$$H_0 : \mu = \mu_0 \text{ against } H_1 : \mu \neq \mu_0 \quad (***)$$

for some $\mu_0 \in \mathbb{R}$.
For $\alpha \in (0,1)$,

$$\varphi : \mathbb{R}^n \ni x \to \begin{cases} 0, & \bar{x} \in \left(\mu_0 - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \mu_0 + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right) \\ 1, & \bar{x} \notin \left(\mu_0 - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \mu_0 + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right) \end{cases}$$

is a level-α test for $(***)$ since

$$P_{\mu_0}(\{x \in \mathbb{R}^n : \varphi(x) = 1\}) = 1 - P_{\mu_0}(\{x \in \mathbb{R}^n : \varphi(x) = 0\})$$

$$= 1 - P_{\mu_0}\left(\left\{x \in \mathbb{R}^n : \bar{x} \in \left(\mu_0 - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \mu_0 + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right)\right\}\right)$$

$$= 1 - P_{\mu_0}\left(\left\{x \in \mathbb{R}^n : \mu_0 \in \left(\bar{x} - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right)\right\}\right)$$

$$= 1 - (1 - \alpha) = \alpha$$

Stochastics KVB - Ex 12

(11)

The delay time T of an atom of a radioactive element is said to have an exponential distribution with parameter λ.

a) Calculate the median of

$$P(X \leq q_{1/2}) \overset{!}{=} \frac{1}{2}$$

$$P(X \leq q_{1/2}) = 1 - e^{-\lambda q_{1/2}} \overset{!}{=} 1/2 \qquad \text{// Strictly increasing}$$

$$\Leftrightarrow \quad e^{-\lambda q_{1/2}} = 1/2$$

$$\Leftrightarrow \quad -\lambda q_{1/2} = \log(1/2)$$

$$\Leftrightarrow q_{1/2} = \frac{\log(1/2)}{-\lambda} \Leftrightarrow q_{0,5} = \frac{\log(2)}{\lambda}$$

b) $P(X \leq q_{1/4}) = 1 - e^{-\lambda q_{1/4}} \overset{!}{=} \frac{1}{4}$

$$\Leftrightarrow \quad e^{-\lambda q_{1/4}} = 3/4$$

$$\Leftrightarrow q_{1/4} = \frac{\log(3/4)}{-\lambda} = \frac{\log(4/3)}{\lambda}$$

$$P(X \leq q_{3/4}) = 1 - e^{-\lambda q_{3/4}} \overset{!}{=} \frac{3}{4}$$

$$\Leftrightarrow e^{-\lambda q_{3/4}} = 1/4$$

$$\Leftrightarrow q_{3/4} = \frac{\log(1/4)}{-\lambda} = \frac{\log(4)}{\lambda}$$

Stochastics - KVB - Exam

①

a) half-life 8 days

$$\beta = \frac{\log(2)}{\lambda} \Leftrightarrow \lambda = \frac{\log(2)}{\beta} \approx 0,0866$$

$$\rightarrow q_{0,25} = \frac{\log(4/3)}{0,0866} \approx 3,322$$

$$\rightarrow q_{0,75} = \frac{\log(4)}{0,0866} \approx 16$$

Stochastics – KLD – Ex 12

(1)

For $\sigma > 0$ let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (P_\mu)_{\mu \in \mathbb{R}})$ be a statistical model, where P_μ denotes the joint distribution of n independent Normal (μ, σ) - distributed random variables $X_1, \dots, X_n$.

a) Determine the smallest possible $c \in \mathbb{R}$ such that the confidence interval C for the expectation μ of the form $C(x) = (\bar{x} - c, \infty)$ is of error level α.

$$P_\mu(\{x \in \mathcal{X} : \mu \in (\bar{x} - c, \infty)\}) \overset{!}{\geq} 1 - \alpha$$

$$= P_\mu(\{x \in \mathcal{X} : \bar{x} \in (\mu - c, \infty)\})$$

$\bar{x} \sim \text{Nord}(\mu, \frac{\sigma}{\sqrt{n}})$

$$= P_\mu\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \in \left(-\frac{\sqrt{n}}{\sigma} c, \infty\right)\right) \quad \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim \text{Nord}(0,1)$$

$$= -\phi\left(-\frac{\sqrt{n}}{\sigma} c\right) + \phi(\infty)$$

$$1 - \alpha \leq P_\mu(\bar{x} - c \leq \mu) = P_\mu\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq \frac{c}{\sigma/\sqrt{n}}\right)$$

$$= \phi\left(\frac{c}{\sigma/\sqrt{n}}\right) = \phi\left(\sqrt{n}\,\frac{c}{\sigma}\right)$$

choose $c \in \mathbb{R}$: $\frac{c}{\sigma} \cdot \sqrt{n} \overset{!}{=} z_{1-\alpha} \rightsquigarrow c = \frac{z_{1-\alpha}}{\sqrt{n}} \cdot \sigma$

$$\Rightarrow C(x) = \left[\bar{x} - \frac{z_{1-\alpha}}{\sqrt{n}} \sigma, \infty\right)$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \sim \text{Nord}(\mu, \frac{\sigma}{\sqrt{n}})$$

- $a \cdot x_1 \sim \text{Nord}(a \cdot \mu, a \cdot \sigma) \quad a \in \mathbb{R} \setminus \{0\}$
- For $y \sim \text{Nord}(m, s)$ ind of x: $x_1 + y \sim \text{Nord}(m + \mu, \sqrt{\sigma^2 + s^2})$

$$\sum_{i=1}^{n} x_i \sim \text{Normal}\left(\mu n, \underbrace{\sqrt{\sigma^2 + \sigma^2 + \dots + \sigma^2}}_{\sqrt{n}\,\sigma}\right)$$

(2)

b) Compare this confidence interval with the one sided Gauss test.

One sided Gauss test: $H_0: \mu \leq \mu_0$

$H_1: \mu > \mu_0$

Decision for H_0 if $\bar{x} \leq \mu_0 - \frac{z_{1-\alpha}}{\sqrt{n}} \sigma$

$\Leftrightarrow \mu_0 \geq \bar{x} - \frac{z_{1-\alpha}}{\sqrt{n}} \sigma$

Stochastik – KVB – 6x12

(3)

One-Sided Gauss test φ of level α with Hypothesis

$H_0 : \mu \leq \mu_0$ and $H_1 : \mu > \mu_0$

a) The actual mean is $\mu_1 > \mu_0$ and let β be a given value for the maximal Type II error for μ_1. Show that the condition $P_{\mu_1}(\{x \in \mathbb{R}^n : \varphi(x) = 0\}) \leq \beta$ is equivalent to

$$n \geq \frac{(z_{1-\alpha} + z_{1-\beta})^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

$$P_{\mu_1}(\{x \in \mathbb{R}^n : \varphi(x) = 0\}) = P_{\mu_1}(\{x \in \mathbb{R}^n : \bar{x} \leq \mu_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}}\})$$

$$= P_{\mu_1}(\{x \in \mathbb{R}^n : \bar{x} - \mu_0 \leq z_{1-\alpha} \frac{\sigma}{\sqrt{n}}\})$$

$$= P_{\mu_1}(\{x \in \mathbb{R}^n : \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \leq z_{1-\alpha}\})$$

Claim: * $\bar{x} \sim N(\mu_1, \sigma/\sqrt{n})$

Proof: $\beta \geq P_{\mu_1}(\{x \in \mathbb{R}^n : \varphi(x) = 0\}) = P_{\mu_1}(\bar{x} \leq \mu_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}})$ (*)

$$= P_{\mu_1}\left(\frac{\bar{x} - \mu_1}{\sigma/\sqrt{n}} \leq \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{1-\alpha}\right) = \phi\left(\frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{1-\alpha}\right)$$

$$\Rightarrow \frac{\mu_0 - \mu_1}{\sigma/\sqrt{n}} + z_{1-\alpha} \leq z_\beta \Leftrightarrow n \geq \frac{(z_{1-\alpha} + z_{1-\beta})^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

∎

Statistics - EX 12

①

In order to reduce 11% of fare evaders, public transport hired more inspectors some time ago. Now they want to test with an error level of 0,05 whether the number of fare evaders has decreased. Therefore, they controlled 457 passengers and caught 35 fare evaders.

a) Suitable statistical Model

$$\left(\{0,1\}^{457}, P(\{0,1\}^{457}), (P_p)_{p\in(0,1)}\right)$$

joint dist of 457 i.i.d $\sim Ber(p)$ r.V.s

b) H_0 & H_1

$H_0: p \geq 11$ $\qquad$ $H_1: p < 11$

Stochastik - KVB - G15

(1)

a) $|\Omega| = 3! = 6$

$\Omega = \{(a,b,c) : c \neq b \neq a,\ c,b,a \in \{1,2,3\}\}$

$P(A) = \frac{|A|}{6}$

(Ω, P)

b) $G = \{\text{Team 1 wins}\}$

$|G| = 2$

$P(G) = \frac{2}{6} = \frac{1}{3}$

c) $R = \{\text{Team 2 is better than at least one of Team 3}\}$

$= \{(2,1,3), (2,3,1), (1,2,3)\}$

$|R| = 3 \quad \Rightarrow P(R) = \frac{1}{2}$

$G \cap R = \{(1,2,3)\} \Rightarrow |G \cap R| = 1$

$P(G \cap R) = \frac{1}{6} = P(G) \cdot P(R) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} \Leftrightarrow$ unabhängig

d) $\underbrace{\text{Team 1 wins}}_{A = G}$ given that $\underbrace{\text{Team 3 is not always the last team}}_{B}$

$$P(G|B) = \frac{P(G \cap B)}{P(B)} = \frac{1/6}{4/6} = \frac{1}{4}$$

$B = \{(3,1,2), (3,2,1), (2,3,1), (1,3,2)\} \Rightarrow |B| = 4$

Stochastik - KW3 - Ex 13

(2)

Let X be a random variable with density

$$f(x) = \begin{cases} 2-2x & , x \in [0,1] \\ 0 & , x \notin [0,1] \end{cases}$$

a) f is density

$f(x) \geq 0$ for $x \notin [0,1]$

for $x \in [0,1]$: $f(x) = 2-2x = 2\underbrace{(1-x)}_{\geq 0} \geq 0$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_0^1 2-2x\,dx = \left[2x - x^2\right]_0^1 = 2-1 = 1$$

$\rightsquigarrow f(x)$ is density

b) Determine the distribution function of X

$$F_X(y) = \int_{-\infty}^{y} f(x)\,dx = \begin{cases} 0 & , x < 0 \\ A & , y \in [0,1] \\ 1 & , y > 1 \end{cases}$$

$$A = \int_{-\infty}^{y} f(x)\,dx = \int_0^y 2-2x\,dx = \left[2x - x^2\right]_0^y = 2y - y^2$$

(23)

c) Compute the expectation and the variance of X

$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx = \int_0^1 x(2-2x)\,dx = \int_0^1 2x - 2x^2\,dx$$

$$= \left[x^2 - \frac{2}{3}x^3\right]_0^1 = 1 - \frac{2}{3} = \frac{1}{3}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_0^1 2x^2 - 2x^3\,dx = \left[\frac{2}{3}x^3 - \frac{1}{2}x^4\right]_0^1$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{4}{6} - \frac{3}{6} = \frac{1}{6}$$

$$\mathrm{Var}[X] = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{6} - \frac{1}{9} = \frac{9}{54} - \frac{6}{54} = \frac{3}{54}$$

d) Show that $m = 1 - \frac{1}{\sqrt{2}}$ is a median of X

$$P(X \le m) = F_X(m) = 2\left(1 - \frac{1}{\sqrt{2}}\right) - \left(1 - \frac{1}{\sqrt{2}}\right)^2$$

$$= 2 - \sqrt{2} - \left(1 - 2/\sqrt{2} + \frac{1}{2}\right) = 2 - \sqrt{2} - 1 + \sqrt{2} - \frac{1}{2}$$

$$= \frac{1}{2}$$

Strictly increasing in this interval $\rightsquigarrow$ median

Statistics KVB - Ex 13

5)

Let X_1 and X_2 be random variables such that X_1 takes values in $\{0,1\}$ and X_2 takes values in $\{-1,0,1\}$. We have

$$P(X_1=0, X_2=i) = \frac{1}{6} \quad \& \quad P(X_1=1, X_2=i) = \frac{1}{4}, \quad i \in \{-1,1\}$$

as well as $P(X_1=0, X_2=0) = \frac{1}{12}$

a)

X_1 \ X_2	-1	0	1	
0	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{5}{12}$
1	$\frac{1}{4}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{7}{12}$
	$\frac{10}{24}$	$\frac{4}{24}$	$\frac{10}{24}$	1

4+6

$$P(X_2=-1) = \frac{10}{24}, \quad P(X_2=0) = \frac{4}{24}, \quad P(X_2=1) = \frac{10}{24}$$

$$P(X_1=0) = \frac{5}{12}, \quad P(X_1=1) = \frac{7}{12}$$

b) $P(X_1=0) = P(X_1=0, X_2=-1) + P(X_1=0, X_2=0), P(X_1=0, X_2=1)$

↗ In Worten aufschreiben

X, Y ind $\Rightarrow \mathrm{Cov}(X,Y)=0$

$\not\Leftarrow$

b) Cov of X_1, X_2

$$\mathrm{Cov}(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2]$$

$$E[X_1] = 1 \cdot P(X_1=1) = \frac{7}{12}$$

$$E[X_2] = -1 \cdot P(X_2=-1) + P(X_2=1) = 0$$

$$X_1 \cdot X_2 \in \{0, -1, 0, 1\}$$

$$E[X_1 X_2] = -1 \cdot \left(\frac{1}{4}\right) + 1 \cdot P\left(\frac{1}{4}\right) = 0$$

$$\mathrm{Cov}(X_1, X_2) = 0$$

Aber nicht unabhängig $P(X_1=1, X_2=1) = \frac{1}{4} \overset{!}{=} \frac{35}{144} \neq P(X_1=1) \cdot P(X_2=1)$

③

Sei Y & Z die davon unabhängigen random Variablen such that $Y \sim \text{Geom}(\frac{1}{2})$ und $Z \sim \text{Ber}(\frac{1}{2})$,

$P(Y=k) = \frac{1}{2^{k+1}}$ für $k \in \mathbb{N}_0$ und $P(Z=1) = P(Z=0) = \frac{1}{2}$

$0 \to \infty$

a) Gesucht $P(Y+Z=\ell)$

$$P(Y+Z=\ell) = \sum_{k=0}^{\infty} P(Y=\ell-k) P(Z=k)$$

$$= \frac{1}{2} P(Y=\ell-1) + \frac{1}{2} P(Y=\ell)$$

$$= \frac{1}{2} \cdot \frac{1}{2^{\ell}} + \frac{1}{2} \frac{1}{2^{\ell+1}} = \frac{1}{2^{\ell+1}} + \frac{1}{2^{\ell+2}} = \frac{2}{2^{\ell+2}} + \frac{1}{2^{\ell+2}}$$

$$= \frac{3}{2^{\ell+2}} \quad , \ell > 0$$

$$= \begin{cases} \frac{1}{2} P(Y=0) = \frac{1}{4} & ; \ell = 0 \\ \frac{3}{2^{\ell+2}} & , \ell > 0 \end{cases}$$

Solution - KVB - EXB

(1)

a)

Let $(X_i)_{i \in \mathbb{N}}$ be a sequence of iid random variables such that $E[X_1^2] < \infty$ and define $S_n = \sum_{i=1}^{n} X_i$ for $n \in \mathbb{N}$

Then

$$\frac{S_n}{n} \xrightarrow{P} E[X_1], \text{ as } n \to \infty$$

b) Prove

$$\lim_{n \to \infty} \sum_{k=0}^{n} \frac{n^k}{k!} e^{-n} = \frac{1}{2}$$

$$S_n = \sum_{k=1}^{n} X_k \sim \text{Poiss}(n), \quad E[S_n] = n$$

$$\text{Var}[S_n] = n$$

$$\frac{S_n - E[S_n]}{\sqrt{\text{Var}[S_n]}} \xrightarrow{d} N \sim \mathcal{N}(0,1)$$

$$P\left(\frac{S_n - E[S_n]}{\sqrt{\text{Var}[S_n]}} \leq t\right) \xrightarrow{n \to \infty} \phi(t)$$

$$\frac{1}{2} = \phi(0) = \lim_{n \to \infty} P\left(\frac{S_n - E[S_n]}{\sqrt{\text{Var}[S_n]}} \leq 0\right)$$

$$= \lim_{n \to \infty} P\left(\frac{S_n - n}{\sqrt{n}} \leq 0\right) = \lim_{n \to \infty} P(S_n \leq n)$$

$$= \lim_{n \to \infty} P(S_n \in \{0, 1, \dots, n\}) = \lim_{n \to \infty} \sum_{k=0}^{n} \frac{n^k}{k!} e^{-n}$$

$\blacksquare$

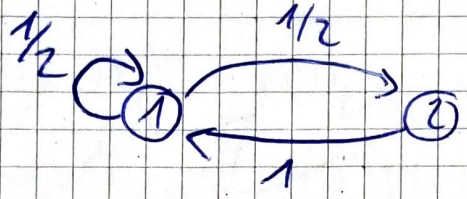
Stochastik – KVD – Gr 13

(5)

a) Let $(X_t)_{t \in \mathbb{N}}$ be a Markov Chain with state space $\{1,2\}$ and transition matrix

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix}$$

$P(X_1 = 1) = \frac{1}{2}$, $P(X_1 = 2) = \frac{1}{2}$



$$p(1) = \left(\frac{1}{2} \quad \frac{1}{2}\right) \qquad p(2) = \left(\frac{1}{2} \quad \frac{1}{2}\right)\begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix} = \left(\frac{3}{4} \quad \frac{1}{4}\right)$$

$$p(1) = p(0) \cdot \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix} = (1/2 \quad 1/2)$$

$\rightsquigarrow p(0) = (1 \quad 0)$

Student - 6x10 - 4VB

①

5) Let $(N_t)_{t \geq 0}$ be a Poisson process with intensity one.
Compute the probability

$$P(N_2 = 4,\ N_{\ln(16)} = 4)$$

$$P(N_2 = 4,\ N_{\ln(16)} = 4) = P(N_2 - N_0 = 4 - 0,\ N_{\ln(16)} - N_2 = 0)$$

independent

$$= P(N_2 - N_0 = 4) \cdot P(N_{\ln(16)} - N_2 = 0)$$

$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$

$$= \frac{(2-0)^4}{4!} e^{-(2-0)} \cdot \frac{(\ln(16) - 2)^0}{0!} e^{-(\ln(16) - 2)}$$

$$= \frac{16}{4} e^{-2} \cdot e^{+2} e^{\log_2(16)^{-1}} = \frac{1}{24}$$

$N_5 - N_t \sim$ Pois $(5 - t)$

$P(N_2 = 4,\ N_{\ln(16)} = 4) = P(N_2 - N_0$

(5)

a) For repeated measurements of an experiment it is known that they are realisations of independent normal random variables with unknown expectation μ and standard deviation $0,25 = \sigma$

One would like to verify that $\mu > 5$.

Provide a suitable statistical model.

$$\left(\mathbb{R}^{25}, \mathcal{B}(\mathbb{R}^{25}), (P_\mu)_{\mu \in \mathbb{R}}\right)$$

$\alpha = 0,05$

$H_0: \mu \leq 5$ vs $H_1: \mu > 5$

$$\varphi: \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \bar{x} < \mu_0 + z_{1-\alpha} \frac{6}{\sqrt{n}} \\ 1, & \bar{x} \geq \mu_0 + z_{1-\alpha} \frac{6_0}{\sqrt{n}} \end{cases}$$

The critical value is $5 + z_{1-0,05} \cdot \frac{0,25}{\sqrt{25}}$

$$= 5,1 + z_{1-0,05} \frac{0,25}{\sqrt{25}} - 0,1$$

$$= 5,082246 \leq \bar{x} = 5,1$$

$\hookrightarrow \varphi(x) = 1 \rightsquigarrow$ reject H_0

(6)

a) Let $(\mathbb{R}, \mathcal{B}(\mathbb{R}), (P_\alpha)_{\alpha>0})$ be thus a statistical model, where P_α is the distribution of a Pareto(α)-distributed rV X.

$$f(x) = \begin{cases} \alpha x^{-\alpha-1} & , x \geq 1 \\ 0 & , x < 1 \end{cases}$$

for $\alpha \geq 0$. Show that $T(x) = \frac{1}{x}$ is an unbiased estimator for $\frac{\alpha}{\alpha+1}$

$$B_{T(x)}\left(\frac{\alpha}{\alpha+1}\right) = E_{\frac{\alpha}{\alpha+1}}[T(x)] - \frac{\alpha}{\alpha+1} = \frac{\alpha}{\alpha+1} - \frac{\alpha}{\alpha+1} = \underline{\underline{0}}$$

$$E_{\frac{\alpha}{\alpha+1}}[T(x)] = \int_{-\infty}^{\infty} x f(x)\,dx = \int_1^\infty \alpha x^{-\alpha}\,dx = \left[\frac{\alpha x^{-\alpha+1}}{-\alpha+1}\right]_1^\infty$$

$$= -\frac{\alpha x^{-\infty}}{\alpha+1} + \frac{\alpha}{\alpha+1}$$

$\hookrightarrow 0$